

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0, 1\}^p$, where $z_j = 1$ indexes a feature in the invariant set S^* (Peters et al., 2016). The posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each environment-specific conditional $\hat{p}_e$. Consistency (Thm 1) requires $\mu_{\min} > 0$, the minimum KL divergence between $\hat{g}$ and any $\hat{p}_e$ over non-invariant features (Wu and Blei, 2025, Def. 1). Contraction (Thm 2) gives $\text{TV}(p(z \mid \mathcal{D}), \delta_{z^*}) = O(2^p e^{-\kappa n E \mu_{\min}})$. Both bounds assume squared loss under a linear model, and the product likelihood weights all environments symmetrically. VI-BIP (Wu and Blei, 2025, Sec. 4) approximates the posterior via mean-field variational inference (Blei et al., 2017), optimizing the ELBO with the U2G gradient estimator.

Critique

Peters et al. (2016), Fan et al. (2024), and Wu and Blei (2025) all enforce the same invariance condition, yet what separates them is what the practitioner assumes about the environments before seeing data. BIP’s symmetric product likelihood weights every environment equally, so its contraction rate $nE \gtrsim O(p/\mu_{\min})$ is bottlenecked by the weakest environment signal. NegDRO makes the opposite assumption: by permitting negative environment weights (Fan et al., 2024, Def. 1), it can amplify a dominant intervention the practitioner knows *a priori* to be strong. These are incompatible design choices, and which is appropriate depends entirely on what the practitioner knows.

Yet no theory characterizes where the boundary lies. There is no decision rule telling a practitioner when symmetric weighting wastes known structure, or when asymmetric weighting overfits to a misidentified dominant environment. The gap between the two frameworks is acknowledged by neither paper.

The experiments do not close this gap. Wu and Blei (2025, Sec. 5) validate VI-BIP exclusively on CRISPR knockouts where $\mu_{\min} > 0$ holds by construction, the one regime where a practitioner *could* justify asymmetric weighting and BIP’s agnosticism is least needed. The empirical case for VI-BIP in observational or low-intervention settings, where symmetric weighting should matter most, remains unmade.

Questions

Both questions below concern whether the practitioner-knowledge gap between BIP and NegDRO can be bridged.

1. A continuous parameter α interpolating between BIP’s symmetric weight constraint and NegDRO’s negative-weight set is straightforward to define. The harder question is whether the minimax estimation rate varies smoothly along this axis, or whether there is a phase transition at some α^* below which symmetric weighting is minimax-optimal and above which asymmetric weighting becomes strictly preferable.
2. Nonparametric extensions of ICP exist for the symmetric case (Heinze-Deml et al., 2018), but no nonparametric analogue of NegDRO’s asymmetric weighting has been proposed. Does the practitioner-knowledge distinction identified above survive when the loss is no longer quadratic, or does the symmetry-versus-asymmetry tradeoff collapse under nonparametric conditionals?

References

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